Tableau Data Visualization Assignment Report

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May 24, 2026

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1 Introduction

This project explores the relationship between state firearm legislation and gun violence in the United States using data spanning 1976–2023. While estimated household gun ownership is included as a contextual variable, it is not treated as the primary explanatory factor. Evidence from countries such as Finland and Switzerland where high gun ownership coexists with low rates of gun violence, suggests that the regulatory environment surrounding firearms may matter more than ownership prevalence alone. This project therefore focuses on whether the strength of state firearm laws is associated with lower rates of firearm homicide (not FSS since that is used as an estimator for estimated gun ownership) and whether changes in legislation over time correspond to measurable shifts in gun violence rates.

1.1 Methodology

The project followed the process below to achieve the goal described above:

1. Dataset selection and join strategy design
2. Data analysis outside Tableau
3. Research questions derived from data exploration
4. Visualization types needed to answer each question and storytell
5. Data source connection, relationships, and calculated fields in Tableau
6. Tableau feature selection (parameters, sets, filters, actions, tooltips)
7. Dashboard design and interactive storytelling
8. Use of AI for further Data Analysis aside from Tableau for more complex visualizations (extra)

1.2 Sources

Numerous sources are used throughout the project such as:

- Official dataset documentation
- Harvard Dataverse and Tufts CTSI dataset metadata
- Academic literature on firearm ownership proxies and gun law effects
- Tableau official documentation
- Lecture notebooks and course materials
- AI assistance (declared below)

Regardless of the source, these are indicated through bibliography and references, or in the case of AI, an explicit note indicating the model and prompt used.

1.3 AI Usage

AI was used during this project in a critical and supervised manner. All AI-generated suggestions were reviewed and scrutinized before being applied, with particular attention to the risk of hallucination. AI was consulted primarily for dataset pairing strategy, question framing, and Tableau workflow guidance, and was not used to generate data or conclusions directly.

2 Datasets

Two public datasets were selected and used in conjunction for this project. Both are accessible through their corresponding links and are available in different formats.

2.1 Dataset 1: Firearm Suicide Proxy for Household Gun Ownership, 1949–2023

The first dataset[1] is a state-year level proxy for household gun ownership in the United States. Direct gun ownership data is not publicly available at the state level, so this dataset uses the firearm suicide proxy methodology: the proportion of suicides committed with a firearm (FSS) is used as an indirect estimate of household gun prevalence by state and year. It is hosted on Harvard Dataverse.

- **Source**: Harvard Dataverse, Kang et al.[2]
- **Grain**: One row per U.S. state per year
- **Key columns**: State, Year, FSS (firearm suicide share), firearm suicides, total suicides, firearm homicide rate, total homicides, population
- **Date span**: 1949–2023 (75 years, 50 states + D.C.)
- **Format**: Comma-separated values (`.csv`)

This dataset was chosen because it provides a long-term, population-adjusted, and academically validated measure of gun prevalence that can be compared across states and over time. FSS is not a literal count of guns per household but an indirect proxy widely used in public health and policy research.[3] However, since FSS is derived from suicide data, it is not used as an outcome variable in this project, doing so would introduce circularity when analysing the effect of gun laws on violence. Instead, FSS is retained as a contextual indicator of estimated gun prevalence, while `firearm_homicide_rate` serves as the primary outcome measure for policy effect analysis.

2.2 Dataset 2: State Firearm Law Database, 1976–2024

The second dataset[4] is maintained by the State Firearm Law Research Group at Tufts University and tracks the presence or absence of 72 firearm safety law provisions across all 50 U.S. states for every year from 1976 to 2024.

- **Source**: Tufts CTSI — State Firearm Law Database[4]
- **Grain**: One row per U.S. state per year
- **Key columns**: State, Year, 72 binary law indicator columns (e.g. permit required, background check, waiting period, stand-your-ground, child access prevention)
- **Date span**: 1976–2024
- **Format**: Excel (`.xlsx`)

This dataset was chosen because it provides a historically accurate, year-by-year record of state firearm legislation, enabling analysis of whether law changes correspond to changes in gun violence rates. It is non-partisan by design and is widely used in academic public health research.[4]

2.3 Join Strategy

Both datasets were joined in Tableau on the shared keys `state` and `year`. The ownership proxy spans 1949–2023 and the law database spans 1976–2024, giving an effective analysis window of **1976–2023**. The join was performed using a Tableau relationship between the two data sources on the matching state-year grain, ensuring the workbook remains reproducible and reviewable.

2.4 Analysis

Prior to connecting the data in Tableau, both files were inspected to understand structure, null patterns, and label consistency. Key observations:

- State names required standardisation across both files
- The ownership proxy file includes several pre-calculated rate columns derived from raw counts and population
- The law database contains 72 binary columns requiring aggregation into a composite law count score via a Tableau calculated field
- The overlapping year range (1976–2023) was confirmed as sufficient for robust longitudinal analysis

2.5 Processing

Key processing steps applied before and during connection to Tableau:

- Standardised state name format across both files to enable the join
- Created a composite **law count** calculated field in Tableau summing all 72 binary law indicator columns per state per year
- Retained population-adjusted rates from the ownership proxy for per-capita analysis
- Removed U.S. territory rows (Puerto Rico, Guam, etc.) to keep the analysis to the 50 states and D.C.

2.6 Research Questions

Three questions were formulated based on data exploration and storytelling intent:

1. Is the combination of high estimated gun ownership and few firearm laws associated with higher firearm homicide and suicide rates?
2. How has the gap in firearm violence rates between high-law and low-law states changed over time?
3. When states increase the number of firearm laws, do firearm homicide and suicide rates change in subsequent years?

2.6.1 Question 1: Ownership, Laws, and Violence

Question 1 establishes the core hypothesis of the project. By plotting estimated gun ownership against law count and encoding gun violence rates visually, the viewer can assess whether the states with the worst outcomes cluster in the high-ownership, low-law quadrant. This is framed as an association analysis rather than a causal claim.

2.6.2 Question 2: High-Law vs. Low-Law States Over Time

Question 2 introduces a temporal dimension. By comparing gun violence trends between states that have progressively added laws and those that have not, the viewer can assess whether the two groups have diverged over time and in which direction.

2.6.3 Question 3: Lagged Effect of Law Changes

Question 3 is the most analytically ambitious. Using the `nextyearfss` column already present in the ownership proxy dataset, it asks whether an increase in law count in a given year is associated with a lower ownership proxy or lower violence rates the following year. This introduces a lagged structure that approximates a policy effect without requiring regression.

3 Visualization Prep

Before building charts in Tableau, design and structural decisions were made to align visualizations with the storytelling intent and assignment requirements.

3.1 Storytelling

The project follows a four-act narrative structure. Act 1 opens with the geographic baseline: where gun violence is highest and what the ownership proxy looks like across states. Act 2 introduces the policy dimension: how many laws each state has passed and how that has changed. Act 3 connects both dimensions in the ownership-laws-violence scatter. Act 4 closes with the policy effect question: do states that added laws show improved outcomes? The viewer moves from observation to exploration to interpretation.

3.2 Colors and Themes

Colors were chosen to encode meaning consistently:

- A **sequential red palette** encodes gun violence rates on maps and scatter plots, where darker shades indicate higher rates.
- A **diverging palette** separates high-law from low-law states in trend and comparison views.
- **Set-based color** encodes user-selected states (highlighted) vs. unselected states (grey) in the interactive line chart.
- Backgrounds are light and gridlines are minimal to reduce visual noise and keep focus on data marks.

3.3 Tableau Feature Choices

The following Tableau features were planned and used across this project:

- **Calculated fields**: Composite law count score (sum of 72 binary columns), law category classification, above/below national average flag, estimated gun-owning households
- **Parameters**: Year selector and Top N selector for ranked bar chart
- **Sets and Combined Sets**: Selected states set for interactive line chart, top/bottom N sets combined for ranked bar chart
- **Filters**: State multiselect, year range
- **Dashboard actions**: Set action (click state to add to selected set and update line chart)
- **Reference lines**: National average in trend views
- **Trend lines**: Linear trend in scatter plot (from Analytics pane)

4 Visualizations

A consistent set of calculated fields and a shared color scheme were applied across all visualizations. Field names are self-explanatory (e.g. `CF_law_count`, `CF_show_state`, `P_year`) to ensure readability and reproducibility.

[redacted]

Figure 1: Q1 – Gun Ownership vs. Law Count, colored by Firearm Homicide Rate

4.1 Q1 – Ownership, Laws, and Violence (Scatter Plot)

A **scatter plot** answers Question 1[2.6.1]. Each point represents one state in a given year. The X axis shows estimated gun ownership (FSS), the Y axis shows the composite law count, and the color encodes the firearm homicide rate. A year parameter allows the viewer to scrub through time and observe how the quadrant distribution has changed.

Insights: States in the high-ownership, low-law quadrant consistently show the darkest color (highest firearm homicide rates). States with high ownership but strong laws perform comparably to low-ownership states, consistent with the international evidence from Finland and Switzerland.

4.2 Q2 – High-Law vs. Low-Law State Trends (Line Chart)

An **interactive line chart** answers Question 2[2.6.2]. The X axis shows year, the Y axis shows firearm homicide rate, and lines are colored by a law tier calculated field (high-law / mid-law / low-law) based on law count tertiles. A Set Action on the dashboard allows the viewer to select specific states for direct comparison.

[redacted]

Figure 2: Q2 – Firearm Violence Trends by Law Tier, 1976–2023

Insights: High-law states show a steeper decline in firearm homicide rates from the 1990s onward compared to low-law states. The gap between tiers has widened over time, particularly after 1993.

4.3 Q3 – Lagged Effect of Law Changes (Dual-Axis Line Chart)

A **dual-axis line chart** answers Question 3[2.6.3]. The primary axis shows firearm homicide rate and the secondary axis shows the composite law count, both over time for a user-selected state. The viewer can observe whether increases in law count precede decreases in violence rates in subsequent years.

Insights: In several states, notable increases in law count are followed by measurable reductions in firearm homicide rates in subsequent years. The relationship is not uniform across all states, and no causal interpretation is claimed.

Figure 3: Q3 – Law Count vs. Firearm Violence Rate Over Time (Dual Axis)

4.3.1 Visualization Summary

Question	Chart	Chart Type	Tableau Feature
Q1[2.6.1]	Fig. 1	Scatter plot	Parameter, calc. field, trend line
Q2[2.6.2]	Fig. 2	Line chart (interactive)	Set, set action, reference line
Q3[2.6.3]	Fig. 3	Dual-axis line chart	Calc. field, filter, tooltip

Table 1: Visualization summary table

Table 1 summarizes:

- 4 chart types used across the project: scatter plot, line chart, dual-axis line chart, and choropleth map (dashboard overview).
- 7 Tableau features used: calculated fields, parameters, sets, combined sets, dashboard set actions, reference lines, and trend lines.
- 2 data sources joined by state and year in Tableau, in two different file formats (`.tab` and `.xlsx`).

5 Conclusions

Visualizations reveal the following:

- States combining high estimated gun ownership with few firearm laws consistently show the highest firearm homicide rates [Fig. 1].
- High-law states have seen steeper declines in firearm violence since the 1990s, widening the gap with low-law states over time [Fig. 2].
- In several states, increases in law count appear to precede reductions in violence rates in subsequent years, though no causal claim is made [Fig. 3].
- Gun ownership alone does not explain violence outcomes — the regulatory context surrounding firearms appears to be the more significant variable, consistent with international evidence.

Limitations include the proxy nature of the FSS ownership estimate, the absence of statistical controls for confounding variables such as poverty and urbanization, and the use of a static join on state name which may mask within-state variation.

6 References

[1] Seokho Kang et al. *Firearm Suicide Proxy for Household Gun Ownership, 1949–2023*. Accessed: 2026-05-18. 2023. DOI: 10.7910/DVN/QVYDUD. URL: https://dataverse.harvard.edu/dataset.xhtml?persistentId=doi:10.7910/DVN/QVYDUD.

[2] Seokho Kang and Taehee Lee. "State-level household gun ownership proxy dataset, 1949–2020". In: *Data in Brief* 50 (2023), p. 109548. DOI: 10.1016/j.dib.2023.109548. URL: https://www.sciencedirect.com/science/article/pii/S2352340923006480.

[3] Seokho Kang et al. "Extending the Firearm Suicide Proxy for Household Gun Ownership". In: *American Journal of Preventive Medicine* (2023). DOI: 10.1016/j.amepre.2023.06.004. URL: https://pubmed.ncbi.nlm.nih.gov/37160211/.

[4] State Firearm Law Research Group. *State Firearm Law Database, 1976–2024*. Accessed: 2026-05-18. Tracks presence or absence of 72 firearm law provisions across all 50 U.S. states, 1976–2024. 2024. URL: https://www.tuftsctsi.org/state-firearm-laws/.